

Lambert Badong

lambertbadong@gmail.com | linkedin.com/in/lambert-badong | lamboprojects.github.io

Education

Simon Fraser University – Bachelor of Applied Science: Mechatronic Systems Engineering, EIT (EGBC) Oct 2024

Experience

Mechanical Design Engineer, FoxFab Power Solutions - Surrey, BC Jul 2025 - Present

- Designed mechanical components and assemblies for switchgear enclosures and control panels in SolidWorks and AutoCAD, including copper bus sizing/bracing and Kirk key interlocking, per GD&T and ASME standards.
- Integrated electrical single-line layouts into mechanical designs for switchgear and control assemblies, ensuring fit, accessibility, and compliance with UL 891 (including Annex G bus bracing), UL 1008, and NEMA standards.
- Developed and maintained Bills of Materials (BOMs), 2D fabrication drawings, and 3D CAD models, ensuring consistency and traceability through design and production.
- Performed root-cause investigations and implemented design improvements to enhance reliability, safety, and standardization across product lines.
- Designed and built the Engineering Tool Hub, a Python desktop app with a native SolidWorks add-in in C#, automating the team's BOM, drawing, and revision workflow and cutting non-design job overhead by 89-94%.

Test Engineer Intern, Verathon Medical (Canada) ULC - Burnaby, BC Sep 2021 - Apr 2022

- Designed and prototyped test fixtures, mechanical enclosures, and validation equipment in SolidWorks to support product verification for bronchoscopes and other regulated medical devices.
- Performed non-destructive testing, including dimensional inspection and functional checks, along with environmental and reliability testing on fixtures and devices.
- Produced over 30 mechanical and technical documents including CAD drawings, BOMs, and verification protocols, ensuring alignment with product requirements.

Electro-Mechanical Tester, Algo Communication Products - Burnaby, BC May - Aug 2021

- Increased operational efficiency by 75% by analyzing manufacturing processes, removing inefficiencies and applying continuous-improvement practices.
- Tested IP paging devices for functional and dimensional compliance using multimeters and test jigs, catching defects before units shipped and reducing customer returns.
- Documented findings in detailed test logs and technical reports that guided product improvements with the engineering team.

Projects

Cane-9 Robotic Navigation Companion September 2023 - August 2024

- Modelled the handle and body in SolidWorks and authored design reports, improving design efficiency by 20% through better system integration.

Selected AI Projects 2025 - Present

- Daybook DMS: offline, encrypted daycare app (Next.js, Electron, SQLite) with billing, BC payroll and compliance.
- Engineering Vault: Obsidian knowledge base run by Claude Code agents; 75+ jobs, 1,600+ linked part notes.
- Kobe Bot: Python Discord bot (~6,200 lines) running household expenses, chores, groceries and bills.

Skills

Electrical: Power Distribution Systems, Single Line Diagrams, Transfer Switches (ATS/MTS), Generator Connection Systems, UL 1008, UL 891, NEMA, CSA

Mechanical: Sheet Metal Design, Design for Manufacturability, Prototyping, Finite Element Analysis, GD&T

Software: SolidWorks, AutoCAD, Bluebeam, Fusion 360, Python, C#, C++, VBA, MATLAB & Simulink

AI Tools: Claude Code, Copilot, ChatGPT, AI agents

Project Management: Design-to-Order Workflows, Cross-Functional Collaboration, Wrike, Jira